QBS 103 Final Project - Submission 1

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Variables that I choose

- **Gene:** CXCR2
- **Continuous covariate:** maternal vitamin D at baseline (`mat_vitD`, ng/mL)
- **Categorical covariate 1:** pregnancy outcome (`preg_preeclampsia`: Normal vs Preeclampsia)
- **Categorical covariate 2:** trimester of blood draw (`collect_trimester`), derived from the continuous variable `sampleCollection_gestAge_wks` (1st trimester = before 14 weeks, 2nd trimester = 14 weeks or later)

Load Packages

```
# load ggplot2 for all plots
library(ggplot2)
```

Load the Data

All files are loaded in their original state, exactly as provided on Canvas.

```
# read the three files
# row.names = 1 uses the first column (probe IDs) as the row names of the expression data
expr <- read.csv(file = 'data/GSE85307_geneExpression.csv', row.names = 1)
covar <- read.csv(file = 'data/GSE85307_covar.csv')
geneNames <- read.csv(file = 'data/GSE85307_geneNames.csv')

# check the dimensions (rows, columns) of each file
dim(expr)
```

```
## [1] 500 157
```

```
dim(covar)
```

```
## [1] 157  10
```

```
dim(geneNames)
```

```
## [1] 750   2
```

```r
# look at the first few rows to see how the data is stored
expr[1:3, 1:3]
```

```
##         GSM2264363 GSM2264379 GSM2264394
## 8089659   6.180152   6.105630    6.31849
## 8083594   4.803436   4.787737    4.80452
## 8035050   9.083982   8.902791    9.23949
```

```r
head(covar)
```

```
##   subjectID      geoID            mat_age_yrs                 preg_preeclampsia
## 1 6053887-M GSM2264346 age_years: 31.06639288 pregnancy_condition: Preeclampsia
## 2 3064425-M GSM2264347 age_years: 27.79466119       pregnancy_condition: Normal
## 3 6094898-M GSM2264348 age_years: 34.31895962       pregnancy_condition: Normal
## 4 7050617-M GSM2264349 age_years: 33.42642026       pregnancy_condition: Normal
## 5 7016015-M GSM2264350 age_years: 28.27926078 pregnancy_condition: Preeclampsia
## 6 7054360-M GSM2264351 age_years: 23.50171116       pregnancy_condition: Normal
##           delivery_gestAge_wks              sampleCollection_gestAge_wks
## 1 gestation_weeks: 37.42857143          blood_draw_at_gestation_week: 10
## 2 gestation_weeks: 39.42857143 blood_draw_at_gestation_week: 17.42857143
## 3 gestation_weeks: 38.85714286          blood_draw_at_gestation_week: 13
## 4 gestation_weeks: 39.14285714 blood_draw_at_gestation_week: 17.28571429
## 5 gestation_weeks: 40.14285714 blood_draw_at_gestation_week: 10.71428571
## 6 gestation_weeks: 41.14285714 blood_draw_at_gestation_week: 10.57142857
##                             batch_Microarray                mat_BMI
## 1 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 37.52407538
## 2 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 20.59144273
## 3 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 20.71118512
## 4 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 31.99358316
## 5 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 27.17734484
## 6 microarray_hybridization_date: 03/19/2013 bmi_kg/m2: 22.28684666
##                                          mat_vitD
## 1 vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94
## 2 vitamind_25(0h)_ng/ml_whole_blood_baseline: 40.3
## 3 vitamind_25(0h)_ng/ml_whole_blood_baseline: 40.8
## 4 vitamind_25(0h)_ng/ml_whole_blood_baseline: 52.2
## 5 vitamind_25(0h)_ng/ml_whole_blood_baseline: 22.8
## 6 vitamind_25(0h)_ng/ml_whole_blood_baseline: 44.6
##                        mat_asthma
## 1 asthma_0=healthy_1=asthmatic: 1
## 2 asthma_0=healthy_1=asthmatic: 1
## 3 asthma_0=healthy_1=asthmatic: 1
## 4 asthma_0=healthy_1=asthmatic: 0
## 5 asthma_0=healthy_1=asthmatic: 0
## 6 asthma_0=healthy_1=asthmatic: 0
```

```r
head(geneNames)
```

```
##   PROBEID      GENE
## 1 8128043      CNR1
## 2 7962489 PLEKHA8P1
## 3 7920271    S100A4
```

```
## 4 7900468      NFYC
## 5 8026496 LINC00661
## 6 7989725     PDCD7
```

Clean the Covariate Data

The first column name has a hidden character at the start of the file, and every covariate value starts with a label (e.g. `"age_years: 31.07"`), so both need to be fixed.

```
# rename the first column (the hidden character at the start of the file changes its name)
colnames(covar)[1] <- 'subjectID'

# remove the label before ": " in every covariate column (columns 3 to 10)
# '.*: ' means "any characters followed by a colon and a space"
for (i in 3:ncol(covar)) {
  covar[, i] <- gsub(pattern = '.*: ', replacement = '', x = covar[, i])
}

# convert the continuous variables from character to numeric
# (they were stored as text because of the labels removed above)
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# convert the categorical variables to factors with readable labels
# the levels argument sets the order (Normal is listed first, so it is the reference group)
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c('Normal', 'Preeclampsia'))
covar$mat_asthma <- factor(ifelse(covar$mat_asthma == 1, 'Asthma', 'Healthy'),
                           levels = c('Healthy', 'Asthma'))
covar$batch_Microarray <- factor(covar$batch_Microarray)

# derive a categorical variable from the continuous gestational age at blood draw
# the first trimester ends at 13 6/7 weeks, so < 14 weeks = 1st trimester
covar$collect_trimester <- factor(ifelse(covar$sampleCollection_gestAge_wks < 14,
                                         '1st trimester (<14 wks)',
                                         '2nd trimester (>=14 wks)'))

# count missing values in the variables I use (all should be 0)
sum(is.na(covar$mat_vitD))
```

```
## [1] 0
```

```
sum(is.na(covar$preg_preeclampsia))
```

```
## [1] 0
```

```
sum(is.na(covar$sampleCollection_gestAge_wks))
```

```
## [1] 0
```

```r
sum(is.na(covar$collect_trimester))
```

```
## [1] 0
```

```r
# check that the conversion worked (no missing values, sensible ranges)
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##
##  delivery_gestAge_wks sampleCollection_gestAge_wks   batch_Microarray
##  Min.   :26.43        Min.   :10.00                03/19/2013:45
##  1st Qu.:38.14        1st Qu.:12.14                03/15/2013:44
##  Median :39.29        Median :13.43                03/22/2013:29
##  Mean   :38.82        Mean   :14.13                03/27/2013:26
##  3rd Qu.:40.00        3rd Qu.:16.43                04/04/2013:10
##  Max.   :42.00        Max.   :18.86                03/18/2013: 1
##                                                    (Other)   : 2
##     mat_BMI         mat_vitD       mat_asthma                collect_trimester
##  Min.   :17.95   Min.   : 5.11   Healthy:94   1st trimester (<14 wks) :87
##  1st Qu.:22.90   1st Qu.:13.40   Asthma :63   2nd trimester (>=14 wks):70
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
##
```

```r
# number of samples in each trimester group
table(covar$collect_trimester)
```

```
##
##  1st trimester (<14 wks) 2nd trimester (>=14 wks)
##                       87                       70
```

Match Probe IDs to Gene Names

```r
# check that every probe in the expression data is in the linker file (should all be TRUE)
table(rownames(expr) %in% geneNames$PROBEID)
```

```
##
## TRUE
##  500
```

```r
# keep only the probes that are in the expression data
geneNames <- geneNames[geneNames$PROBEID %in% rownames(expr), ]

# put both in the same order by probe id so that row i matches row i
expr <- expr[order(as.numeric(rownames(expr))), ]
geneNames <- geneNames[order(geneNames$PROBEID), ]

# check that the probe ids are now in the same order (should all be true)
table(rownames(expr) == geneNames$PROBEID)
```

```
##
## TRUE
##  500
```

```r
# replace the probe ids with gene names
rownames(expr) <- geneNames$GENE
expr[1:3, 1:3]
```

```
##         GSM2264363 GSM2264379 GSM2264394
## PHF13     7.197650   7.298383   7.147849
## TNFRSF8   8.094866   8.049821   7.929165
## CTRC      6.504210   6.674909   6.446666
```

Match Gene Expression to Covariates

The columns of the expression file are **not** in the same order as the rows of the covariate file, so the samples need to be put in the same order before combining.

```r
# check that the same samples are in both files (should all be TRUE in both directions)
table(colnames(expr) %in% covar$geoID)
```

```
##
## TRUE
##  157
```

```r
table(covar$geoID %in% colnames(expr))
```

```
##
## TRUE
##  157
```

```r
# check whether they are already in the same order (they are not, so sorting is needed)
table(colnames(expr) == covar$geoID)
```

```
##
## FALSE
##   157
```

```r
# flip the expression data so that rows are samples and columns are genes
exprT <- as.data.frame(t(expr))

# put both data frames in the same order by sample id
exprT <- exprT[order(rownames(exprT)), ]
covar <- covar[order(covar$geoID), ]

# check that the sample ids are now in the same order (should all be true)
table(rownames(exprT) == covar$geoID)
```

```
##
## TRUE
##  157
```

```r
# combine the covariates and gene expression into one data frame
dat <- data.frame(covar, exprT)

# double check one sample: the value in the combined data should match the original file
dat$CXCR2[dat$geoID == 'GSM2264363'] == expr['CXCR2', 'GSM2264363']
```

```
## [1] TRUE
```

```r
# final sample size (rows = samples, columns = covariates + genes)
dim(dat)
```

```
## [1] 157 511
```

All 157 samples were matched, so no samples were removed.

Summary Table

```r
# continuous variables: mean and standard deviation
mean(dat$CXCR2)
```

```
## [1] 12.17055
```

```r
sd(dat$CXCR2)
```

```
## [1] 0.2040686
```

```r
mean(dat$mat_vitD)
```

```
## [1] 22.98255
```

```r
sd(dat$mat_vitD)
```

```
## [1] 13.14992
```

```r
# categorical variables: counts, then percentages (rounded to 1 decimal place)
table(dat$preg_preeclampsia)
```

```
##
##       Normal Preeclampsia
##          110           47
```

```r
round(prop.table(table(dat$preg_preeclampsia)) * 100, 1)
```

```
##
##       Normal Preeclampsia
##         70.1         29.9
```

```r
table(dat$collect_trimester)
```

```
##
##  1st trimester (<14 wks) 2nd trimester (>=14 wks)
##                       87                       70
```

```r
round(prop.table(table(dat$collect_trimester)) * 100, 1)
```

```
##
##  1st trimester (<14 wks) 2nd trimester (>=14 wks)
##                    55.4                     44.6
```

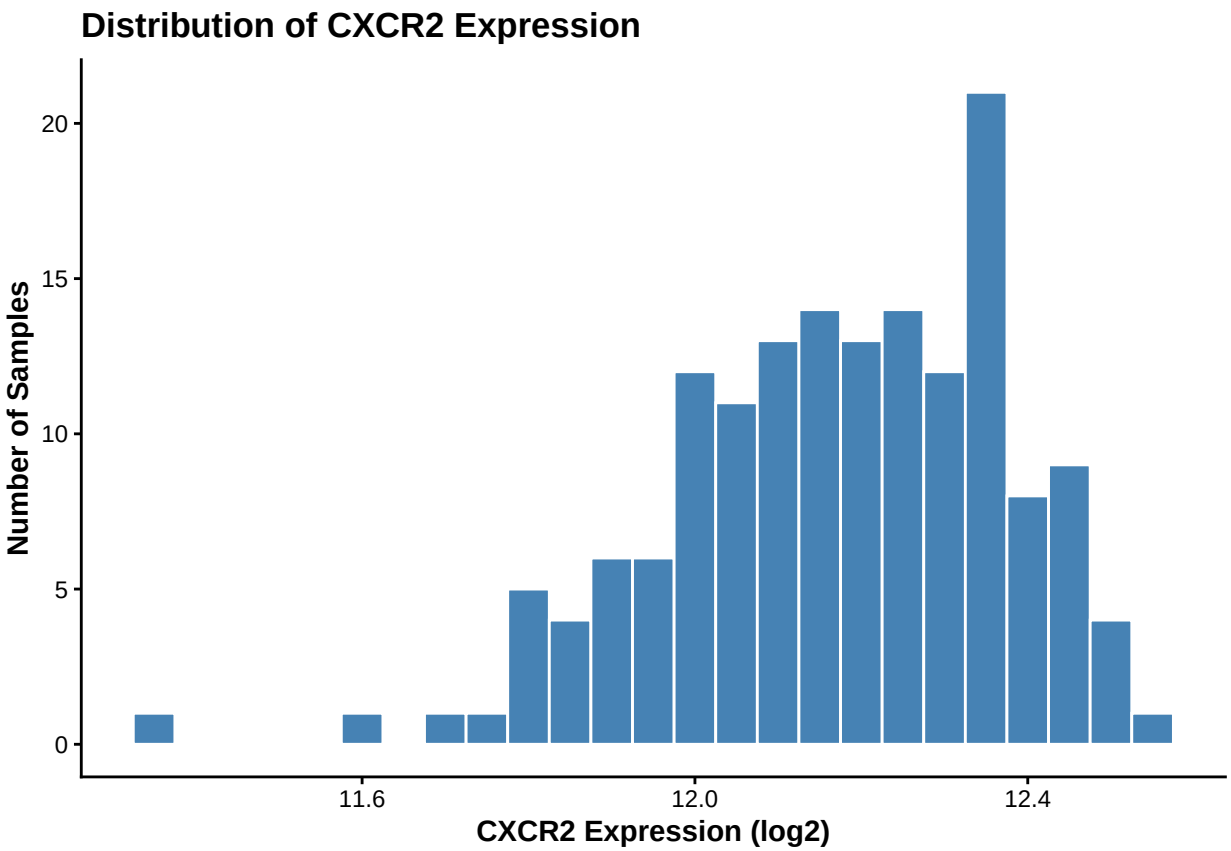
```r
# number of samples in each boxplot group (rows = trimester, columns = pregnancy outcome)
table(dat$collect_trimester, dat$preg_preeclampsia)
```

```
##
##                            Normal Preeclampsia
##   1st trimester (<14 wks)      58           29
##   2nd trimester (>=14 wks)     52           18
```

Plots

```r
# histogram of cxcr2 expression
# binwidth = 0.05 (log2 units) is small enough to show the shape of the distribution
ggplot(dat, aes(x = CXCR2)) +
  geom_histogram(binwidth = 0.05, fill = 'steelblue', color = 'white') +
  labs(title = 'Distribution of CXCR2 Expression',
       x = 'CXCR2 Expression (log2)',
       y = 'Number of Samples') +
  theme_classic() +
  theme(plot.title = element_text(face = 'bold'),
        axis.title = element_text(face = 'bold'))
```

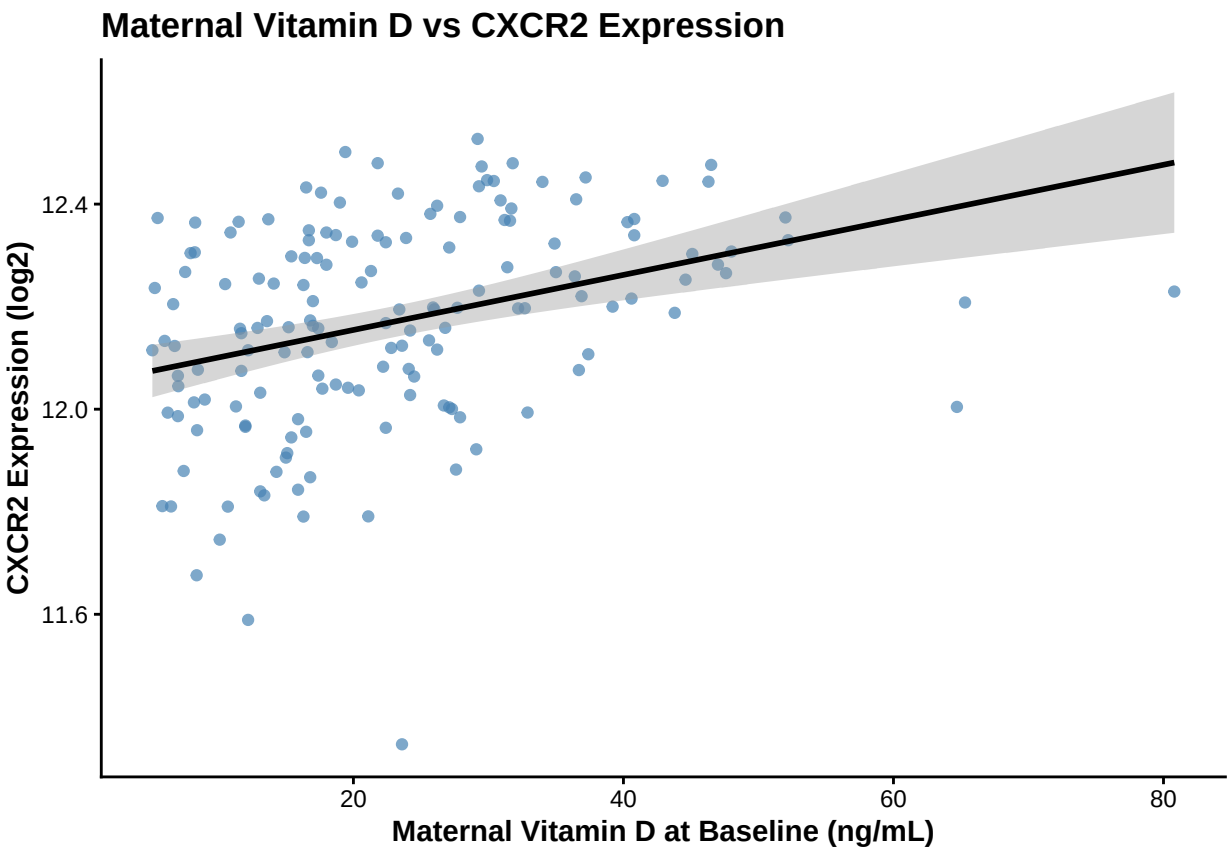
Histogram of CXCR2 expression



CXCR2 expression is centered around 12.2 (log2) and is slightly left-skewed. Most samples fall between about 11.8 and 12.5, with a few low values near 11.4 to 11.7.

```
# scatterplot of maternal vitamin d vs cxcr2 expression with a linear trend line
# alpha = 0.7 makes the points slightly transparent so overlapping points are visible
# formula = y ~ x tells geom_smooth which model to fit (a straight line)
ggplot(dat, aes(x = mat_vitD, y = CXCR2)) +
  geom_point(color = 'steelblue', alpha = 0.7) +
  geom_smooth(method = 'lm', formula = y ~ x, color = 'black') +
  labs(title = 'Maternal Vitamin D vs CXCR2 Expression',
       x = 'Maternal Vitamin D at Baseline (ng/mL)',
       y = 'CXCR2 Expression (log2)') +
  theme_classic() +
  theme(plot.title = element_text(face = 'bold'),
        axis.title = element_text(face = 'bold'))
```

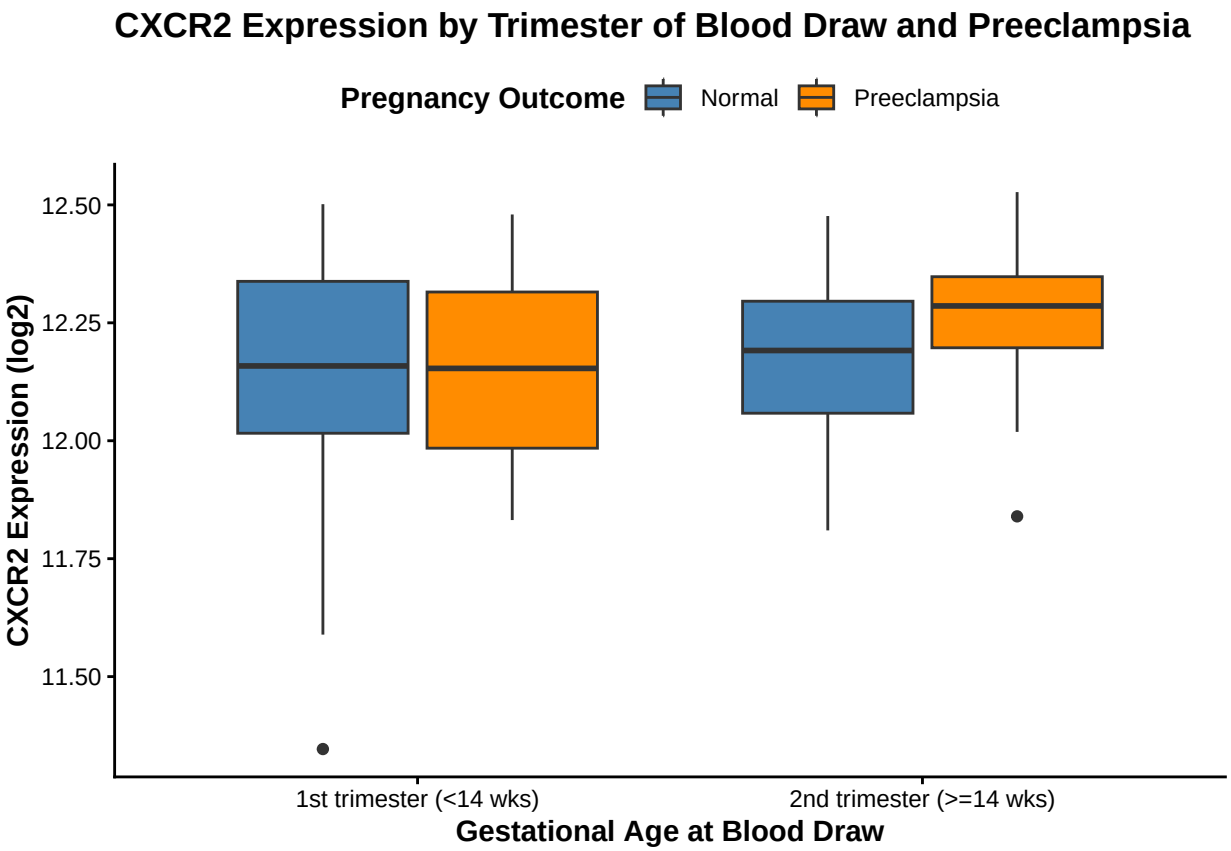
Scatterplot of maternal vitamin D vs CXCR2 expression



The trend line rises slightly, so higher maternal vitamin D is weakly associated with higher CXCR2 expression. The points are widely scattered around the line, so the relationship is weak. The two samples above 60 ng/mL widen the confidence band on the right side.

```
# boxplot of cxcr2 expression by trimester of blood draw and pregnancy outcome
# x = trimester, fill = pregnancy outcome, so each trimester has one box per outcome
# (the number of samples in each box is in the Summary Table section above)
ggplot(dat, aes(x = collect_trimester, y = CXCR2, fill = preg_preeclampsia)) +
  geom_boxplot() +
  scale_fill_manual(values = c('steelblue', 'darkorange')) +
  labs(title = 'CXCR2 Expression by Trimester of Blood Draw and Preeclampsia',
       x = 'Gestational Age at Blood Draw',
       y = 'CXCR2 Expression (log2)',
       fill = 'Pregnancy Outcome') +
  theme_classic() +
  theme(plot.title = element_text(face = 'bold'),
        axis.title = element_text(face = 'bold'),
        legend.title = element_text(face = 'bold'),
        legend.position = 'top')
```

Boxplot of CXCR2 expression by trimester and pregnancy outcome



In the 1st trimester, the Normal and Preeclampsia medians are nearly identical (about 12.15). In the 2nd trimester, the Preeclampsia median (about 12.29) appears higher than the Normal median (about 12.19). The 2nd-trimester Preeclampsia group has only 18 samples (the other groups have 58, 29, and 52), so this difference should be interpreted with caution. No statistical test was run, so these are descriptive observations only.